

GOODLAND, KS, USA

WMO: 724650

Lat: 39.367N

Lon: 101.693W

Elev: 3656

StdP: 12.86

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23065

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.9	(c) 6.2	(d) -6.0	(e) 4.6	(f) 5.3	(g) -1.2	(h) 5.9	(i) 12.2	(j) 35.3	(k) 39.1	(l) 31.9	(m) 38.8	(n) 10.9	(o) 340	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.4	(c) 97.7	(d) 66.6	(e) 94.8	(f) 66.8	(g) 92.0	(h) 66.5	(i) 71.5	(j) 86.9	(k) 70.3	(l) 85.9	(m) 69.1	(n) 84.8	(o) 12.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	115.3	76.3	66.0	109.7	75.2	64.6	104.7	74.1	37.7	86.8	36.5	85.7	35.4	84.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.0	27.2	24.3	-8.0	101.9	4.6	2.7	-11.3	103.9	-13.9	105.5	-16.5	107.0	-19.8	108.9		
			-8.5	73.9	4.4	1.6	-11.7	75.1	-14.3	76.0	-16.8	76.9	-20.0	78.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	30.6	32.4	41.4	49.3	59.5	70.8	76.4	74.0	65.5	52.2	40.3	31.0
	DBStd	18.82	11.11	11.71	10.94	9.68	8.74	7.18	5.57	5.36	8.54	9.73	10.23	10.66
	HDD50	2553	603	498	302	127	20	1	0	0	6	89	317	590
	HDD65	5734	1067	914	730	474	214	29	3	5	96	407	741	1054
	CDD50	3303	2	5	37	107	314	623	817	743	471	158	25	1
	CDD65	1006	0	0	0	3	43	201	355	283	111	10	0	0
	CDH74	12817	1	6	57	197	784	2567	4268	3116	1510	294	17	0
	CDH80	6247	0	1	9	46	290	1249	2336	1556	671	89	0	0
Wind	WSAvg	11.9	11.7	12.4	13.1	14.1	12.9	12.1	10.8	10.3	11.2	11.8	11.7	11.5
Precipitation	PrecAvg	18.4	0.4	0.4	1.0	1.5	3.1	3.2	2.7	2.2	1.4	1.2	0.6	0.5
	PrecMax	29.6	1.6	1.6	3.6	4.8	8.2	9.4	8.0	6.0	5.4	4.3	2.1	2.8
	PrecMin	9.2	0.0	0.0	0.0	0.0	0.5	0.1	0.4	0.1	0.0	0.0	0.0	0.0
	PrecStd	4.4	0.3	0.4	0.9	1.0	1.8	2.0	1.6	1.6	1.3	1.1	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.3	72.1	80.6	85.3	92.7	100.7	101.3	99.3	95.8	88.2	76.4	65.9
		MCWB	45.3	47.7	52.9	55.3	61.7	64.8	67.1	66.4	63.5	58.1	51.0	46.4
	2%	DB	60.5	64.0	73.9	79.8	87.4	95.4	98.4	95.3	91.8	81.9	70.8	59.3
		MCWB	42.8	44.7	49.7	54.3	60.3	65.4	67.4	67.1	63.7	56.5	49.0	42.7
	5%	DB	53.8	57.7	68.0	74.7	82.6	91.6	95.6	92.5	87.8	76.3	64.5	53.2
		MCWB	40.0	42.2	48.1	53.1	60.3	65.4	67.9	67.2	63.3	55.3	47.1	39.9
	10%	DB	47.6	50.9	61.8	68.8	77.9	87.6	92.6	89.2	83.4	70.4	57.9	47.5
		MCWB	37.0	38.7	45.8	51.4	58.9	65.2	68.0	67.1	62.0	53.1	44.2	37.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.0	49.4	56.0	59.2	67.2	71.8	73.4	72.9	69.1	62.9	54.0	47.2
		MCDB	64.3	67.4	72.7	74.8	82.2	86.5	88.4	87.5	84.8	76.7	68.4	62.8
	2%	WB	43.4	45.6	52.3	56.9	64.4	69.6	71.9	71.2	67.3	59.8	50.7	43.6
		MCDB	59.6	62.5	68.0	72.6	78.9	85.0	87.8	86.0	83.2	72.8	67.2	57.6
	5%	WB	40.6	42.7	49.3	54.6	62.4	68.2	70.7	69.9	65.7	57.4	48.0	40.6
		MCDB	53.1	56.5	65.8	70.4	76.9	84.4	86.9	84.8	80.7	72.2	63.2	52.9
	10%	WB	37.5	39.4	46.4	52.3	60.3	66.7	69.4	68.7	64.0	54.9	45.0	37.2
		MCDB	47.2	50.4	61.1	67.2	74.3	82.5	85.8	83.5	78.3	67.7	57.6	47.0
Mean Daily Temperature Range	5% DB	MDBR	25.3	25.1	28.3	27.4	26.4	27.5	27.4	26.3	27.8	27.5	26.7	24.8
		MCDBR	34.3	35.5	38.4	37.9	34.0	33.6	32.2	31.6	33.4	35.9	36.5	33.8
		MCWBR	20.0	20.0	19.2	17.6	13.8	11.3	9.6	9.9	11.5	15.8	19.1	20.2
	5% WB	MCDBR	32.4	33.9	35.1	32.6	29.5	29.0	27.8	26.7	28.9	30.7	34.5	32.3
		MCWBR	19.6	19.8	19.0	16.6	13.9	11.7	10.1	10.2	11.5	15.2	18.6	19.8
Clear-Sky Solar Irradiance	taub		0.251	0.259	0.286	0.314	0.339	0.357	0.375	0.358	0.328	0.288	0.259	0.250
	taud		2.568	2.559	2.490	2.436	2.432	2.412	2.407	2.417	2.490	2.575	2.579	2.561
	Ebn at Noon		301	312	310	305	298	291	285	287	292	296	295	293
	Edh at Noon		22	26	31	35	36	37	37	36	31	25	22	21
All-Sky Solar Radiation	RadAvg		808	1066	1471	1804	2083	2333	2318	2022	1655	1207	882	702
	RadStd		56	82	117	89	126	101	82	99	89	110	61	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page